

Divyanshu Kumar

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LinkedIn — GitHub — Portfolio

AI & Software Engineer
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SUMMARY

AI & Software Engineer with 1 year of experience building end-to-end applications using Python, Flask, REST APIs, and real-time system integration. Skilled in SDLC, debugging, testing, deployment, and Agile team collaboration. Experienced in applying Machine Learning, Computer Vision, and RAG-based NLP solutions to automation and production-focused use cases.

EDUCATION

- Dumka Engineering College** Jharkhand, India
B.Tech in Computer Science Engineering — CGPA: 8.55 Graduated 2025

EXPERIENCE

- Tech Genius Institute** Gandhinagar, India
AI & Robotics Engineer / Mentor August 2025 – Present
 - Delivered STEM, AI, ML and Robotics education while mentoring students on competition-oriented engineering projects, resulting in 15+ deployable prototypes.
 - Managed documentation, testing workflows, prototyping cycles and project execution aligned with Agile planning.
 - Collaborated with cross-functional teams to deliver reliable embedded + AI solutions involving hardware integration and software deployment.
- Gyantu** Remote
AI Intern Nov 2024 – Jan 2025
 - Built a context-aware chatbot using LLaMA 3.2, LangChain and ChromaDB, improving user query resolution and engagement.
 - Developed a real-time Streamlit interface for interactive RAG-powered responses, enhancing user retention by 40%.
- TechInterex** Remote
Data Science Intern Feb 2025 – Apr 2025
 - Built ML pipelines involving preprocessing, feature engineering, experimentation and dashboard visualizations.
 - Worked across training projects such as recommendation engines and predictive modeling using Pandas and Scikit-learn.

PROJECTS

- Gene Ontology-Based Classification (Research Project):**
 - Conducted a research-oriented study on gene classification using Gene Ontology (GO) features and machine learning techniques.
 - Designed preprocessing pipelines for high-dimensional biological data, including feature encoding, filtering, and experimental evaluation.
 - Implemented and fine-tuned TabTransformer-based deep learning models, achieving approximately 10% accuracy improvement compared to traditional machine learning approaches.
 - Explored Positive-Unlabeled (PU) learning strategies for handling highly imbalanced biological datasets.
 - Research paper currently in verification stage; not yet published.
 - Tech: Python, Pandas, NumPy, Scikit-learn, TabTransformer, Deep Learning.
- AI-Powered In-Car Assistant (Driver Monitoring System):**
 - Developed a real-time driver monitoring system detecting drowsiness, emotion and road-rage using CNN, 3D-CNN, and dlib-based EAR analysis.
 - Integrated NodeMCU + ultrasonic/IR sensors with Flask server and MQTT for real-time vehicle control and live demo prototype.
 - Built hardware-software pipeline using Python, OpenCV, TensorFlow, Streamlit, NodeMCU, L298N.
- Conversational AI Chatbot (RAG + LLaMA 3.2):**
 - Built RAG-based contextual chatbot using LLaMA 3.2, LangChain and ChromaDB for document-aware response generation.
 - Implemented Streamlit UI for real-time interaction and searchable document Q&A improving response relevance and usability.
 - Tech: Python, LangChain, Ollama, RAG, Streamlit.

SKILLS SUMMARY

- Languages:** Python, C++, C, SQL
- Software Engineering:** Flask, REST APIs, Git, Linux, Docker basics, Unit Testing
- Machine Learning & CV:** TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, Transformers
- Tools:** Streamlit, Jupyter, Selenium, BeautifulSoup, MQTT, NodeMCU
- Other:** SDLC, Agile, Documentation, Deployment & Debugging

ACHIEVEMENTS

- : Winner — Innovathon (BIT Sindri): Eyeficient — Smart Grocery Quality Control System
- : Finalist — HackFest '24 (IIT ISM): Unified Customer Insights
- : Awarded Best Final Year Project: Driver Assistance System
- : Top 3 Rank in CSE Department & 250+ DSA problems solved